Worksheet 3.2 — Graphs of Rational Functions

Grade 12 Advanced Functions (MHF4U) · Unit 3: Rational Functions

The Big Ideas

Sketch from the skeleton: intercepts, vertical and horizontal asymptotes, and the behaviour on each side of a VA.

- x-intercept: numerator $= 0$. y-intercept: evaluate $f(0)$.
- VA: denominator $= 0$. HA: from the degrees.
- Near a VA the curve rushes to $\pm\infty$ test each side.

Core Concepts

Locate asymptotes first

Find vertical asymptotes from the denominator's zeros, and the end-behaviour asymptote by comparing the degrees of top and bottom.

Intercepts and holes

Numerator zeros give x-intercepts; a factor shared by top and bottom creates a **hole** rather than an asymptote.

Sketching

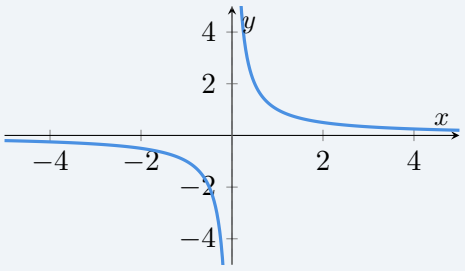
Plot the intercepts, draw the asymptotes as guides, and check the sign in each region to finish the graph.

Worked Examples

Example 1 — The reciprocal

Sketch $y = \dfrac{1}{x}$.

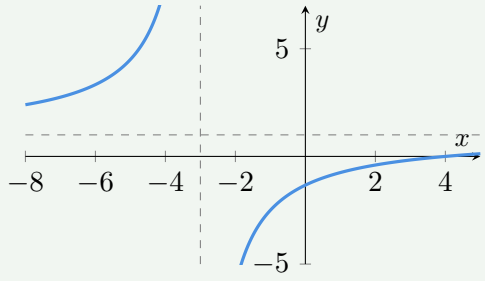
Solution. VA $x = 0$, HA $y = 0$, no intercepts:



Example 2 — Full skeleton

For $y = \dfrac{x-4}{x+3}$, find intercepts and asymptotes.

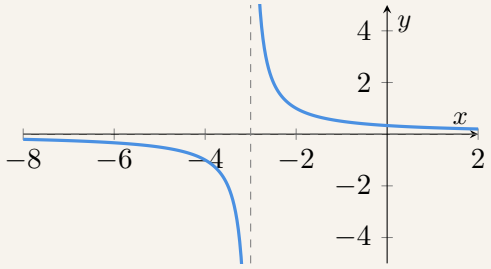
Solution. x-int $(4, 0)$; y-int $f(0) = -\frac{4}{3}$; VA $x = -3$; HA $y = 1$:



Example 3 — Behaviour near a VA

How does $y = \dfrac{1}{x+3}$ behave near $x = -3$?

Solution. Just left, $y \to -\infty$; just right, $y \to +\infty$:



Example 4 — y-intercept

y-intercept of $y = \dfrac{x+3}{x-1}$?

Solution. $f(0) = \dfrac{3}{-1} = -3$, so $(0, -3)$.

Example 5 — End behaviour

As $x \to \infty$, $y = \dfrac{2x+1}{x-3} \to$?

Solution. Equal degrees $\frac{2}{1}$, so $y \to 2$.

Example 6 — VA and HA

VA and HA of $y = \dfrac{2}{x-6}$?

Solution. VA $x = 6$, HA $y = 0$.

Example 7 — x-intercept

x-intercept of $y = \dfrac{x-8}{x+3}$?

Solution. Numerator 0 at $x = 8$: $(8, 0)$.

Example 8 — y-intercept

y-intercept of $y = \dfrac{x+6}{x-3}$?

Solution. $f(0) = \dfrac{6}{-3} = -2$: $(0, -2)$.

Example 9 — Equal degrees

HA of $y = \dfrac{6x}{x-5}$?

Solution. $y = 6$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

VA and HA of $y = \dfrac{3}{x+4}$?

Question 2

x-intercept of $y = \dfrac{x-7}{x+1}$?

Question 3

y-intercept of $y = \dfrac{x+9}{x-3}$?

Question 4

HA of $y = \dfrac{8x}{x+6}$?

Question 5

As $x \to \infty$, $y = \dfrac{3x+1}{2x-5} \to$?

Question 6

VA of $y = \dfrac{x}{x-3}$?

Question 7

x-intercept of $y = \dfrac{2x-6}{x+4}$?

Question 8

y-intercept of $y = \dfrac{x-1}{x+5}$?

Question 9

HA of $y = \dfrac{4x}{x-2}$?

Question 10

Behaviour of $y = \dfrac{1}{x+1}$ just right of $x = -1$?

Question 11

x- and y-intercepts of $y = \dfrac{x-3}{x+1}$?

Question 12

As $x \to \infty$, $y = \dfrac{5x-2}{x+1} \to$?

Question 13 — Challenge

Find all intercepts and asymptotes of $y = \dfrac{x-1}{x+2}$, then sketch.

Question Recap & Answer Key

Question Recap

1. VA and HA of $y = \dfrac{3}{x+4}$?
2. x-intercept of $y = \dfrac{x-7}{x+1}$?
3. y-intercept of $y = \dfrac{x+9}{x-3}$?
4. HA of $y = \dfrac{8x}{x+6}$?
5. As $x \to \infty$, $y = \dfrac{3x+1}{2x-5} \to$?
6. VA of $y = \dfrac{x}{x-3}$?
7. x-intercept of $y = \dfrac{2x-6}{x+4}$?
8. y-intercept of $y = \dfrac{x-1}{x+5}$?
9. HA of $y = \dfrac{4x}{x-2}$?
10. Behaviour of $y = \dfrac{1}{x+1}$ just right of $x = -1$?
11. x- and y-intercepts of $y = \dfrac{x-3}{x+1}$?
12. As $x \to \infty$, $y = \dfrac{5x-2}{x+1} \to$?
13. Find all intercepts and asymptotes of $y = \dfrac{x-1}{x+2}$, then sketch.

✓ Answer Key

1. VA $x = -4$, HA $y = 0$
2. $(7, 0)$
3. $(0, -3)$
4. $y = 8$
5. $y = \frac{3}{2}$
6. $x = 3$
7. $(3, 0)$
8. $(0, -\frac{1}{5})$
9. $y = 4$
10. $y \to +\infty$
11. $(3, 0)$ and $(0, -3)$
12. $y = 5$
13. x-int $(1, 0)$, y-int $(0, -\frac{1}{2})$, VA $x = -2$, HA $y = 1$